

Muditha Fernando

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 Google Scholar  Homepage

"A self-motivated individual equipped with strong fundamental knowledge and has experience in solving real-world problems in Computer Vision and Machine Learning."

RESEARCH INTERESTS

- Computer Vision
- Self-Supervised Learning
- Robustness
- Multi Modal Learning

EDUCATION

University of Moratuwa, Sri Lanka

Jan. 2020 – June 2024

B.Sc.(Hons) Electronic and Telecommunication Engineering

CGPA: 3.94/4.20

Dean's list: Semester 1,2,6,7,8

Highlighted Courses: Machine Vision, Advances in Machine Vision, Bio-Medical Image Processing, Fundamentals of Image Processing and Machine Vision, Pattern Recognition and Machine Intelligence, Applied Statistics, Linear Algebra, Calculus, Signals and Systems, Random Signal Processing, Data Structures and Algorithms.

EXPERIENCE

Computer Vision Research Group, University of Moratuwa, Sri Lanka

Aug 2024 – Present

Research Student

- Self-supervised video-representation learning
- Language guided semi-supervised domain generalization (ICPR 2026)

Pekoe Pte Ltd, Singapore (Remote)

Jun 2024 – Present

Senior AI Engineer

- Researched the use of non-contrastive self-supervised learning on tea image datasets.
- Investigated supervised contrastive learning approaches for robust representation learning.
- Research on statistical feature extraction methods and texture extraction for tea images.
- Studied the applicability of masked image modeling methods to tea image datasets.
- Research on the transferability of foundational models for tea image datasets.

Pekoe Pte Ltd, Singapore (Remote)

Sept 2023 – Mar 2024

Part-time AI Engineering Intern

- Conducted research on image-based methods for assessing tea sample homogeneity.
- Investigated multi-task learning approaches for feature extraction from tea images.

Pekoe Pte Ltd, Singapore (Remote)

Jan 2023 – Jul 2023

AI Engineering Intern

- Applied self-supervised learning techniques for representation learning using CNN and ViT architectures.
- Investigated the use of Generative Adversarial Networks (GANs) for synthetic data generation.
- Developed time series forecasting models using SARIMAX and XGBoost.

PUBLICATIONS AND THESIS

- **M. Fernando**, K. Kailainathan, K. Nagarathnam, I. U. B. Senavirathne, & R. Rodrigo, "SemAlign: Language Guided Semi-supervised Domain Generalization". In Proceedings of the International Conference on Pattern Recognition (ICPR 2026).
- **M. Fernando***, A. Perera*, T. Rajaka*, B. Yasarathna*, R. Rodrigo, S. Ramasinghe, "Perception Module for World Model" (2024). [Thesis Report]
Joint first authorship indicated by *.

RESEARCH PROJECTS

Semi Supervised Domain Generalization

Jan 2025 - Dec 2025

- Aligning image representations with class label embeddings for semi-supervised domain generalization, and introduced associated loss constraints.
- Introduced a computationally efficient Fourier amplitude swapping-based data augmentation method.
- Achieves SOTA performance across multiple datasets
- Outcome: [Paper][Code]

Perception Module for World Models

Aug 2023 – Apr 2024

Undergraduate Thesis Project - University of Moratuwa

- Designed a shared encoder model to jointly learn content, flow, and depth features from unlabeled monocular videos using a self supervised multi task framework.
- Implemented task-specific decoders to capture spatial and temporal dependencies and improve multi-task feature consistency.

PROJECTS

A Study on Contrastive and Classification Objectives for Representation Learning

- This project compares contrastive and classification objectives for representation learning in low labeled data settings. The objective is to investigate the learnt representation quality of these methods. [Code]

SKILLS

Languages: English (Professional), Sinhala (Native)

Programming Languages: Python, MATLAB, C++, System Verilog

Developer Tools: Git, VS Code, Confluence, JIRA

Libraries: PyTorch, Pandas, NumPy, Matplotlib, OpenCV, Scikit learn, ITK

Softwares: SolidWorks, Proteus, Altium, Multisim, Webots, Coppeliasim

COMPETITIONS

IEEEEXtreme 15.0: 24-hour Programming Competition

2021

Secured 24th place country wide and 332nd global

E-Yantra robotics competition IIT Bombay

2021

Secured 4th place in berryminator category

SERVICE

Teaching : Lab instructor for EN 1094 Laboratory Practical course

Mentoring: Mentor for computer vision competition teams at University of Moratuwa, Sri Lanka

REFERENCES

Dr. Ranga Rodrigo

B.Sc. Eng. Hons. (Moratuwa), M.E.Sc. (Western),
Ph.D. (Western), MIET, MIEEE

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